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Pseudorabies virus for treating tumors

Abstract

A pseudorabies virus (PRV) or a modified form thereof, or a genome sequence or a cDNA sequence containing the PRV or the modified form thereof, or nucleic acid molecules of a complementary sequence of the cDNA sequence, for treating tumors of subjects and/or reducing or inhibiting tumor recurrence, and for preparation of a pharmaceutical composition used for treating the tumors of the subjects and/or reducing or inhibiting the tumor recurrence. A method for treating tumors and/or reducing or inhibiting tumor recurrence, comprising a step of administering, on a subject having a need, the PRV or the modified form thereof, or nucleic acid molecules of the genome sequence containing the PRV or the modified form thereof.

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Background/Summary

REFERENCE TO A SEQUENCE LISTING

(1) In accordance with 37 CFR § 1.52(e)(5), the present specification makes reference to a Sequence Listing submitted electronically as a .txt file named “531837US_ST25.txt”. The .txt file

was generated on Aug.17, 2020 and is 352 kb in size. The entire contents of the Sequence Listing are herein incorporated by reference.

TECHNICAL FIELD

(2) The invention relates to the fields of virus and tumor therapy. In particular, the present invention relates to use of a pseudorabies virus (PRV) or modified form thereof, or a nucleic acid molecule comprising a genomic sequence of the PRV or modified form thereof, for treating a tumor and/or reducing or inhibiting tumor recurrence in a subject (e.g., a human), and use thereof in manufacture of a medicament for treating a tumor and/or reducing or inhibiting tumor recurrence in a subject (e.g., a human). The present invention also relates to a method for treating a tumor and/or reducing or inhibiting tumor recurrence, which comprises a step of administering to a subject in need thereof a PRV or modified form thereof, or a nucleic acid molecule comprising a genomic sequence of the PRV or modified form thereof.

BACKGROUND ART

(3) The current means for treatment of malignant tumors mainly include surgical treatment, chemotherapy and radiotherapy. These traditional therapies are not satisfactory in the treatment of metastasized tumors, and may further cause great harm to the health of patients.

(4) An oncolytic virus is a virus that can replicate itself in tumor cells, thereby killing or lysing tumor cells, or stopping the growth of tumor cells. When in vivo treatment is performed, oncolytic viruses exhibit specific selectivity for tumor cells and can directly induce the death of tumor cells, but have little or no effect on normal cells; meanwhile, oncolytic viruses can also stimulate the response of B lymphocytes and T lymphocytes in immune system, thereby indirectly killing tumor cells. Therefore, oncolytic viruses are considered to be a promising tumor treatment method.

(5) Pseudorabies virus (PRV), also known as porcine herpesvirus type I, infectious bulbar paralysis virus, “mad itch” virus, Aujeszky's disease virus, belongs to the genus varicellovirus of subfamily α -herpesvirus. It is a herpesvirus that causes fever, severe itching (except for pigs) and encephalomyelitis as the main symptoms in a variety of domestic animals and wild animals such as cattle, sheep, pig, dog and cat. Pig is a natural host of pseudorabies virus, and exhibits symptoms of acute infectious diseases such as fever, diarrhea, dyspnea, encephalomyelitis and reproductive disorders after infection. Pseudorabies virus has the following characteristics: first, as a double-stranded DNA virus, there is not a process of integrating its genome into the host chromosome in a host; second, although pseudorabies virus has been endemic in some lower mammals, it has not been reported by official literature that it can infect human and cause disease; further, pseudorabies virus does not contain an oncogene. At present, pseudorabies virus is widely used as a veterinary vaccine, etc., but there is no report about pseudorabies virus having oncolytic activity in the art.

CONTENTS OF THE INVENTION

(6) In the present invention, unless otherwise stated, the scientific and technical terms used herein have the meaning commonly understood by those skilled in the art. In addition, the operating steps for cell culture, virology, biochemistry, cell biology, nucleic acid chemistry and so on used herein are conventional steps widely used in the corresponding fields. Meanwhile, in order to better understand the present invention, definitions and explanations of related terms are provided below.

(7) As used herein, the term “pseudorabies virus (PRV)” refers to a virus of the genus varicellovirus (Varicellovirus) of subfamily α -herpesvirus (Alphaherpesvirinae). The genome of the virus is a linear double-stranded DNA with a size of 130 kd to 150 kd. The genome of the virus includes a unique long region sequence (UL) and a unique short region sequence (US), and there are a terminal repeat sequence (TRR) and internal repeat sequence (IRS) on both sides of the US. The genes of PRV are named according to their regions and the order of discovery, but they can also be named by the proteins encoded by them. The genes encoding structural proteins include US2 (28K), US3 (PK), US4 (gG), US6 (gD), US7 (gI), US8 (gE), US9 (11K) and other genes in the US region, and capsid protein genes of UL9 (OBP), UL27 (gB), UL (gH), UL (TK), UL (gC), DNA polymerase gene and other genes in the UL region. The non-structural protein genes located in the

UL region include UL54 which participates in transcriptional regulation. In the present invention, the expression “pseudorabies virus (PRV)” refers to a wild-type PRV, which can be isolated from a natural source and has not been intentionally modified by man, and the examples thereof include the wild strains Bartha-K61 (BK61) and HB-98, as well as various isolates from specimens (for example, the isolate described in Example 1 of the present invention). The wild-type PRV genome sequence is well known in the art, and can be found in various public databases (for example, GenBank database accession number: JF797217.1).

(8) As used herein, the term “oncolytic virus” refers to a virus that can infect a tumor cell, replicate in the tumor cell, and cause the death and lysis of the tumor cell, or prevent the growth of the tumor cell. Preferably, the virus has minimal toxic effects on a non-tumor cell.

(9) As used herein, the term “tumor-specificity” refers to selectively exhibiting a biological function or activity in a tumor cell. For example, in the present invention, when the term “tumor specificity” is used to describe the killing selectivity of a virus, it means that the virus can selectively kill a tumor cell without killing or hardly killing a non-tumor cell, or, the virus is more effective in killing a tumor cell than killing a non-tumor cell.

(10) As used herein, the term “oncolytic activity” mainly comprises tumor-killing activity. When describing the oncolytic activity of a virus, the oncolytic activity of the virus can typically be measured by indicators such as its ability to infect a tumor cell, its ability to replicate in the tumor cell, and/or its ability to kill the tumor cell. The oncolytic activity of a virus can be measured using any method known in the art. For example, the ability of a virus to infect a tumor cell can be evaluated by measuring the viral dose required to infect a given percentage of tumor cells (e.g., 50% of cells); the ability to replicate in a tumor cell can be evaluated by measuring the growth of the virus in the tumor cell; the ability to kill a tumor cell can be evaluated by monitoring cytopathic effect (CPE) or measuring tumor cell activity.

(11) As used herein, the term “modified form” of a virus refers to a modified virus obtained by modifying a wild-type virus, which retains the desired activity (e.g., oncolytic activity) of the wild-type virus. In the present invention, “modified form” of PRV includes, but is not limited to, a modified PRV virus, which has a substitution, insertion or deletion of one or more nucleotides compared to the genomic sequence of a wild-type PRV, and at least retains the oncolytic activity of PRV. It should be understood that the modified form of PRV or the modified PRV of the present invention is not limited by the method for production. For example, the modified form of PRV or the modified PRV of the present invention can be produced by homologous recombination, or can be prepared by culturing a host cell infected with the modified form or the modified PRV.

(12) As used herein, the term “EP0 protein” refers to the early protein 0 of the PRV virus, which is encoded by the EP0 gene and is a transcriptional activator expressed early by the PRV virus. The amino acid sequence of the EP0 protein is known and can be found in, for example, public databases (for example, EM64001.1).

(13) As used herein, the expression “do not express a functional EP0 protein” means that when a virus or viral genome infects a cell, the virus or viral genome cannot produce or express an EP0 protein with biological function or activity. For example, the virus or viral genome may not produce or express EP0 protein at all due to gene deletion, or may produce or express EP0 protein that has no biological function and activity due to a loss-of-function mutation.

(14) As used herein, the term “loss-of-function mutation” refers to a mutation that causes the protein encoded and expressed by the mutant gene to lose its biological function and activity. Loss-of-function mutation includes, but is not limited to, missense mutation, nonsense mutation, frameshift mutation, base deletion, base substitution, base addition, and any combination thereof (e.g., deletion or substitution or addition of a gene fragment), as long as the gene containing the loss-of-function mutation cannot produce or express a protein with biological function or activity.

(15) As used herein, the expression “cDNA sequence of PRV” refers to a DNA sequence obtained by reverse transcription using mRNA transcribed from the viral genome as a template, which

differs from the genome sequence only in that the cDNA sequence does not contain the intron sequence in the genomic sequence.

(16) As used herein, the term “exogenous nucleotide sequence” refers to an artificially introduced nucleotide sequence that is foreign to the original sequence. Exogenous nucleotide sequence includes, but is not limited to, any genes or nucleotide sequences not found in the viral genome. However, in some cases, it is preferred that the exogenous nucleotide sequence encodes a polypeptide having a therapeutic use, such as an immunomodulatory polypeptide, cytokine, chemokine, anti-tumor protein or polypeptide, and the like.

(17) As used herein, the term “immunomodulatory polypeptide” refers to a polypeptide capable of modulating the function of an immune cell, examples thereof include, but are not limited to, CD40L, OX40L, inducible costimulatory molecule (ICOS), FTL3L, LIGHT, CD137L, CD70, 4-1BB, GITR, and CD28 (see, for example, Khalil D N, Smith E L, Brentjens R J, et al. The future of cancer treatment: immunomodulation, CARs and combination immunotherapy [J]. *Nat Rev Clin Oncol*, 2016, 13 (5): 273-290).

(18) As used herein, the term “cytokine” has a meaning well known to those skilled in the art. However, in the present invention, when the oncolytic virus of the present invention is used to treat a tumor, it is particularly preferred that the cytokine is a cytokine that can be used for tumor treatment. Examples of “cytokine” include, but are not limited to, interleukins (e.g., IL-2, IL-12 and IL-15), interferons (e.g., IFN α , IFN β , IFN γ), tumor necrosis factors (e.g., TNF α), colony stimulating factors (e.g., GM-CSF), and any combination thereof (see, for example, Ardolino M, Hsu J, Raulet D H. Cytokine treatment in cancer immunotherapy [J]. *Oncotarget*, 2015, 6 (23): 19346-19347).

(19) As used herein, the term “chemokine” has a meaning well known to those skilled in the art. However, in the method of the present invention, when the oncolytic virus of the present invention is used to treat a tumor, it is particularly preferred that the cytokine is a chemokine that can be used for tumor treatment. Examples of “chemokine” include, but are not limited to, CCL2, RANTES, CCL7, CCL9, CCL10, CCL12, CCL15, CCL19, CCL21, CCL20, XCL-1, and any combination thereof (Homey B, Muller A, Zlotnik A. CHEMOKINES: AGENTS FOR THE IMMUNOTHERAPY OF CANCER [J]. *Nat Rev Immunol*, 2002, 2: 175-184).

(20) As used herein, the term “anti-tumor protein or polypeptide” refers to a protein or polypeptide that has tumor therapeutic activity, including but not limited to: (1) a protein or polypeptide that is toxic to a cell, can inhibit cell proliferation, or induce apoptosis, its examples include but are not limited to, thymidine kinase TK (TK/GCV), TRAIL, and FasL (see, for example, Candolfi M, King G D, Muhammad A G, et al. Evaluation of proapoptotic transgenes to use in combination with Flt3L in an immune-stimulatory gene therapy approach for Glioblastoma multiforme (GBM) [J]. *FASEB J*, 2008, 22: 1077.13); (2) a protein or polypeptide with immunotherapeutic effect, such as immune checkpoint inhibitor, its examples include but are not limited to, anti-PD-1 antibody, anti-PD-L1 antibody, anti-TIGIT antibody, anti-BTLA antibody, anti-CTLA-4 antibody, anti-Tim-3 antibody, anti-Lag-3 antibody, anti-CD137 antibody, anti-OX40 antibody, anti-GITR antibody, anti-CD73 antibody, anti-KIR antibody, anti-ICOS antibody, anti-CSF1R antibody (see, for example, Nolan E, Savas P, Policheni A N, et al. Combined immune checkpoint blockade as a therapeutic strategy for BRCA1-mutated breast cancer [J]. *Science Trans Med*, 2017, 9: eaa14922; all of which are incorporated herein by reference); (3) a tumor-specific targeting antibody, its examples include but are not limited to, anti-HER2 antibody (e.g., Herceptin), anti-CD20 antibody (e.g., Rituximab), anti-VEGF antibody (e.g., Bevacizumab), anti-EGFR antibody (e.g., Cetuximab), or any combination thereof; (4) a protein or polypeptide that inhibits tumor angiogenesis, its examples include but are not limited to, anti-vascular endothelial growth factor (anti-VEGF) single chain antibody (scFv), VEGF-derived polypeptide (e.g., .sub.D(LPR), KSRVRKKGKGQKRKRKKSRYK, etc.), and ATN-161 (see, for example, Rosca E V, Koskimaki J E, Rivera C G, et al. Anti-angiogenic peptides for cancer therapeutics [J]. *Curr Pharm Biotechnol*, 2011, 12 (8): 1101-1116; all of which are

incorporated herein by reference).

(21) As used herein, the term “identity” refers to the match degree between two polypeptides or between two nucleic acids. When two sequences for comparison have the same monomer sub-unit of base or amino acid at a certain site (e.g., each of two DNA molecules has an adenine at a certain site, or each of two proteins/polypeptides has a lysine at a certain site), the two molecules are identical at the site. The percent identity between two sequences is a function of the number of identical sites shared by the two sequences over the total number of sites for comparison $\times 100$. For example, if 6 of 10 sites of two sequences are matched, these two sequences have an identity of 60%. For example, DNA sequences: CTGACT and CAGGTT share an identity of 50% (3 of 6 sites are matched). Generally, the comparison of two sequences is conducted in a manner to produce maximum identity. Such alignment can be conducted by for example using a computer program such as Align program (DNASTar, Inc.) which is based on the method of Needleman, et al. (J. Mol. Biol. 48:443-453, 1970). The percentage of identity between two amino acid sequences can also be determined using the algorithm of E. Meyers and W. Miller (Comput. Appl. Biosci., 4:11-17 (1988)) which has been incorporated into the ALIGN program (version 2.0), using a PAM120 weight residue table, and with a gap length penalty of 12 and a gap penalty of 4. In addition, the percentage of identity between two amino acid sequences can be determined by the algorithm of Needleman and Wunsch (J. Mol. Biol. 48:444-453 (1970)) which has been incorporated into the GAP program in the GCG software package (available at World Wide Web.gcg.com), using either a Blossum 62 matrix or a PAM250 matrix, and with a gap weight of 16, 14, 12, 10, 8, 6, or 4 and a length weight of 1, 2, 3, 4, 5, or 6.

(22) As used herein, the term “vector” refers to a nucleic acid delivery vehicle into which a polynucleotide can be inserted. When a vector enables expression of a protein encoded by an inserted polynucleotide, the vector is called an expression vector. The vector can be introduced into a host cell by transformation, transduction or transfection, so that a genetic material element carried thereby can be expressed in the host cell. Vectors are well known to those skilled in the art and include, but are not limited to: plasmids; phagemids; cosmids; artificial chromosomes, such as yeast artificial chromosomes (YAC), bacterial artificial chromosomes (BAC), or P1 derived artificial chromosomes (PAC); phages such as λ -phage or M13 phage and animal viruses. Animal viruses that can be used as vectors include, but are not limited to, retroviruses (including lentiviruses), adenoviruses, adeno-associated viruses, herpesviruses (such as herpes simplex virus), poxviruses, baculoviruses, papillomaviruses, and papovaviruses (such as SV40). A vector may contain a variety of elements that control expression, including, but not limited to, promoter sequences, transcription initiation sequences, enhancer sequences, elements for selection, and reporter genes. In addition, the vector may contain a replication initiation site.

(23) As used herein, the term “promoter” has the meaning well known to those skilled in the art, and refers to a non-coding nucleotide sequence located in the upstream of a gene that can activate the expression of a downstream gene.

(24) As used herein, the term “pharmaceutically acceptable carrier and/or excipient” refers to a carrier and/or excipient that is pharmacologically and/or physiologically compatible with the subject and the active ingredient, which is well known in the art (see, for example, Remington's Pharmaceutical Sciences. Edited by Gennaro A R, 19th ed. Pennsylvania: Mack Publishing Company, 1995), and includes, but is not limited to: pH adjusting agents, surfactants, ionic strength enhancers, agents to maintain osmotic pressure, agents to delay absorption, diluents, adjuvants, preservatives, stabilizers, etc. For example, pH adjusting agents include, but are not limited to, phosphate buffered saline. Surfactants include, but are not limited to, cationic, anionic or non-ionic surfactants, such as Tween-80. Ionic strength enhancers include, but are not limited to, sodium chloride. Agents that maintain osmotic pressure include, but are not limited to, sugar, NaCl, and the like. Agents that delay absorption include, but are not limited to, monostearate and gelatin. Diluents include, but are not limited to, water, aqueous buffers (such as buffered saline), alcohols and

polyols (such as glycerol), and the like. Adjuvants include, but are not limited to, aluminum adjuvants (such as aluminum hydroxide), Freund's adjuvants (such as complete Freund's adjuvant), and the like. Preservatives include, but are not limited to, various antibacterial and antifungal agents, such as thimerosal, 2-phenoxyethanol, parabens, trichloro-t-butanol, phenol, sorbic acid, and the like. Stabilizers have the meaning commonly understood by those skilled in the art, which can stabilize the desired activity (such as oncolytic activity) of the active ingredients in the drug, including but not limited to sodium glutamate, gelatin, SPGA, sugars (e.g., sorbitol, mannitol, starch, sucrose, lactose, dextran, or glucose), amino acids (e.g., glutamic acid, glycine), proteins (e.g., dried whey, albumin, or casein) or their degradation products (e.g., lactalbumin hydrolysates).

(25) As used herein, the term "treatment" refers to treating or curing a disease (e.g., tumor), delaying the onset of a symptom of disease (e.g., tumor), and/or delaying the development of a disease (e.g., tumor).

(26) As used herein, the term "effective amount" refers to an amount that can effectively achieve the intended purpose. For example, a therapeutically effective amount may be an amount that is effective or sufficient to treat or cure a disease (e.g., tumor), delay the onset of a symptom of disease (e.g., tumor), and/or delay the development of a disease (e.g., tumor). Such an effective amount can be easily determined by a person skilled in the art or a doctor, and can be related to the intended purpose (e.g., treatment), the general health, age, gender, body weight of subject, the severity of disease to be treated, complications, administration routes, etc. The determination of such effective amount is completely within the ability of those skilled in the art.

(27) As used herein, the term "subject" refers to a mammal, such as a primate mammal, such as a human. In certain embodiments, the subject (e.g., human) has a tumor, or is at risk of having a tumor.

(28) After extensive experiments and repeated explorations, the inventors of the present application have unexpectedly discovered that pseudorabies virus (PRV) has a broad-spectrum and significant tumor cell killing ability. Based on this discovery, the inventors have developed a new oncolytic virus for treating a tumor and a method for tumor treatment based on the virus.

(29) Therefore, in a first aspect, the present invention provides a use of a pseudorabies virus (PRV) or modified form thereof or a nucleic acid molecule for treating a tumor and/or reducing or inhibiting tumor recurrence in a subject, or for the manufacture of a medicament for treating a tumor and/or reducing or inhibiting tumor recurrence in a subject; wherein the nucleic acid molecule comprises a sequence selected from the following: (1) a genomic sequence or cDNA sequence of a PRV or modified form thereof; (2) a complementary sequence of the cDNA sequence.

(30) In certain preferred embodiments, the PRV is a wild-type PRV. In certain preferred embodiments, the PRV may be a strain isolated from an animal infected with a pseudorabies virus (PRV).

(31) In certain preferred embodiments, the genomic sequence of the PRV or modified form thereof has a sequence identity of at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% to the nucleotide sequence as shown in SEQ ID NO: 1. In certain preferred embodiments, the genomic sequence of the PRV or modified form thereof is the nucleotide sequence as shown in SEQ ID NO: 1.

(32) In certain preferred embodiments, the modified form is a modified PRV, which has substitution, insertion or deletion of one or more nucleotides in its genome as compared to a wild-type PRV.

(33) In certain preferred embodiments, the modified PRV has one or more modifications selected from the following as compared to a wild-type PRV: (1) a deletion or mutation of one or more endogenous genes; (2) a mutation, deletion or insertion of one or more nucleotides in an untranslated region (e.g., promoter); (3) an insertion of one or more exogenous nucleotide

sequences; and (4) any combination of the above three items.

(34) In certain preferred embodiments, the modified PRV does not express a functional EP0 protein. As well known to those skilled in the art, the gene encoding the EP0 protein (EP0 gene) can be modified to prevent the functional expression of the protein. For example, a loss-of-function mutation can be introduced into the gene encoding the EP0 protein, or the gene encoding the EP0 protein can be deleted or replaced with an exogenous nucleotide sequence (e.g., a nucleotide sequence encoding an exogenous protein) to prevent the functional expression of the target protein.

(35) Therefore, in certain preferred embodiments, the genome of the modified PRV of the present invention is modified as follows: the EP0 gene contains a loss-of-function mutation (e.g., addition, deletion and/or substitution of one or more nucleotides), or is deleted or replaced with an exogenous nucleotide sequence (e.g., a nucleotide sequence encoding an exogenous protein). In certain preferred embodiments, the loss-of-function mutation is selected from missense mutations, nonsense mutation, frameshift mutation, base deletion, base substitution, base addition, and any combination thereof (e.g., a deletion or substitution or addition of a gene fragment).

(36) In certain preferred embodiments, the EP0 gene is deleted. In such embodiments, the modified PRV does not express EP0 protein. In certain preferred embodiments, the EP0 gene is replaced with an exogenous nucleotide sequence (e.g., a nucleotide sequence encoding an exogenous protein). In certain exemplary embodiments, the exogenous protein is a fluorescent protein.

(37) In certain preferred embodiments, the genomic sequence of the modified PRV has a sequence identity of at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% to the nucleotide sequence as shown in SEQ ID NO: 4. In certain preferred embodiments, the genomic sequence of the modified PRV is shown in SEQ ID NO: 4.

(38) In certain preferred embodiments, the modified PRV comprises a mutation, deletion or insertion of one or more nucleotides located in a non-coding region (e.g., a promoter).

(39) In certain preferred embodiments, the genome of the modified PRV comprises the following modification: an original promoter of one or more PRV genes is replaced with a tumor-specific promoter. In the present invention, the original promoter of any PRV gene can be replaced with a tumor-specific promoter. In certain preferred embodiments, the tumor-specific promoter is selected from the group consisting of human telomerase reverse transcriptase (hTERT) promoter, alpha-fetoprotein promoter, carcinoembryonic antigen promoter, prostate-specific antigen promoter, Survivin promoter and cyclooxygenase COX-2 promoter.

(40) In the present invention, the use of tumor-specific promoter is advantageous under certain circumstances, for example, it is conducive to improve the tumor specificity of an oncolytic virus. In normal cells, the activity of tumor-specific promoter is usually at a low level, while in tumor cells its activity is at a high level. Therefore, the use of tumor-specific promoter to regulate the transcription expression of some key proteins of PRV can allow the virus to obtain tumor-specific killing ability. Therefore, in some cases, replacing the original promoter of a key protein of PRV with a tumor-specific promoter can reduce the virus's ability to kill normal cells, thereby reducing toxic and side effects without affecting its ability to kill tumor cells.

(41) In certain preferred embodiments, the modified PRV comprises an exogenous nucleotide sequence.

(42) In certain preferred embodiments, the exogenous nucleotide sequence encodes an exogenous protein selected from the group consisting of fluorescent protein, immunomodulatory polypeptide, cytokine, chemokine, anti-tumor protein or polypeptide.

(43) In certain preferred embodiments, the fluorescent protein is selected from the group consisting of green fluorescent protein, red fluorescent protein, blue fluorescent protein, yellow fluorescent protein, and any combination thereof.

(44) In certain preferred embodiments, the immunomodulatory polypeptide is selected from the group consisting of CD40L, OX40L, inducible costimulatory molecules (ICOS), FTL3L, LIGHT,

CD137L, CD70, 4-1BB, GITR, CD28, and any combination thereof.

(45) In certain preferred embodiments, the cytokine is selected from the group consisting of interleukins (e.g., IL-2, IL-12 and IL-15), interferons (e.g., IFN α , IFN β , IFN γ), tumor necrosis factors (e.g., TNF α), colony stimulating factors (e.g., GM-CSF), and any combination thereof.

(46) In certain preferred embodiments, the chemokine is selected from the group consisting of CCL2, RANTES, CCL7, CCL9, CCL10, CCL12, CCL15, CCL19, CCL21, CCL20, XCL-1, and any combination thereof.

(47) In certain preferred embodiments, the anti-tumor protein or polypeptide is selected from the group consisting of: cytotoxic peptides, such as thymidine kinase TK (TK/GCV), TRAIL, FasL, or any combination thereof; immune checkpoint inhibitors, such as PD-1 antibody, anti-PD-L1 antibody, anti-TIGIT antibody, anti-BTLA antibody, anti-CTLA-4 antibody, anti-Tim-3 antibody, anti-Lag-3 antibody, anti-CD137 antibody, anti-OX40 antibody, anti-GITR antibody, anti-CD73 antibody, anti-KIR antibody, anti-ICOS antibody, anti-CSF1R antibody, or any combination thereof; tumor-specific targeting antibodies, such as anti-HER2 antibody (e.g., Herceptin), anti-CD20 antibody (e.g., Rituximab), anti-VEGF antibody (e.g., Bevacizumab), anti-EGFR antibody (e.g., Cetuximab), or any combination thereof; proteins or polypeptides that inhibit tumor angiogenesis, such as anti-vascular endothelial growth factor (anti-VEGF) antibody, VEGF-derived polypeptides (e.g., .sub.D(LPR), KSRVRKGKGQQRKRKRKKSRYK, etc.), ATN-161 or any combination thereof; and any combination of the above proteins or polypeptides.

(48) In certain preferred embodiments, the modified PRV comprises at least one insertion of the exogenous nucleotide sequence as described above and/or at least one mutation, deletion or insertion in the untranslated region (e.g., promoter) as described above.

(49) In certain preferred embodiments, the modified PRV does not express a functional EP0 protein and comprises at least one insertion of an exogenous nucleotide sequence as described above and/or at least one mutation, deletion or insertion in the untranslated region (e.g., promoter) as described above.

(50) In the present application, the modified PRV of the present invention can be obtained by a technique well known in the art. For example, a loss-of-function mutation can be introduced into the viral gene through deletion, substitution or insertion of bases to inactivate the viral gene functionally. In certain exemplary embodiments, the viral gene is functionally inactivated by deletion (e.g., deletion of the entire gene or part thereof). In such embodiments, at least 25%, at least 50%, at least 75%, or 100% of the viral gene sequence of interest may be deleted, or at least 10 bp, at least 100 bp, or at least 1000 bp of the viral gene sequence of interest may be deleted. In certain exemplary embodiments, a frameshift mutation may be introduced by insertion or deletion of bases, thereby inactivating the viral gene functionally. In certain exemplary embodiments, the viral gene is functionally inactivated by replacing the entire gene of interest or a part thereof with an exogenous nucleotide sequence.

(51) In certain preferred embodiments, the modified PRV of the present invention can be obtained by CRISPR/Cas9 technology. The CRISPR/Cas9 technology is known in the art, for example, see Ran F A, Feng Zhang et al. Nature, 2013, 2281-2308; which is incorporated herein by reference in its entirety. In such embodiments, the viral genome of wild-type PRV is typically modified (e.g., by insertion of an exogenous nucleotide sequence, deletion or mutation of an endogenous gene, or mutation in an untranslated region) to obtain the modified PRV.

(52) The PRV or modified form thereof according to the present invention may be pretreated to reduce or eliminate the immune response against the virus in a subject, wherein the pretreatment may include: packaging the PRV in liposomes or micelles, and/or removing the viral capsid protein using a protease (e.g., chymotrypsin or trypsin) to reduce humoral and/or cellular immunity to the virus in a host.

(53) In the present invention, the PRV or modified form thereof as described herein can be serially passaged for adaptation in tumor cells in tumor cells. In certain preferred embodiments, the tumor

cells may be tumor cell lines or tumor cell stains known in the art, or may be tumor cells surgically excised or clinically isolated from the body of an individual (e.g., a subject) with a tumor. In certain preferred embodiments, the PRV or modified form thereof is serially passaged for adaptation in tumor cells obtained from an individual (e.g., a subject) having a tumor. In certain preferred embodiments, the tumor cells are surgically excised or clinically isolated from the body of an individual (e.g., a subject) with a tumor. In certain preferred embodiments, the method for serial passaging for adaptation comprises a plurality of (e.g., at least 5, at least 10, at least 15, at least 20) cycles consisting of the following processes: 1) infecting a target tumor cell with the virus; 2) harvesting the virus in a supernatant; and 3) reinfecting a fresh target tumor cell with the obtained virus.

(54) In certain preferred embodiments, the PRV and modified form thereof as described above may be used in combination. Therefore, the medicament may comprise one or more of the PRV and modified forms thereof.

(55) In certain preferred embodiments, the nucleic acid molecule consists of a genomic sequence or cDNA sequence of the PRV or modified form thereof as described herein, or a complementary sequence of the cDNA sequence.

(56) In certain preferred embodiments, the nucleic acid molecule has a genomic sequence of the PRV or modified form thereof as described herein. In certain preferred embodiments, the nucleic acid molecule has a nucleotide sequence selected from the following: (1) a nucleotide sequence as shown in SEQ ID NO: 1 or 4; (2) a nucleotide sequence having a sequence identity of at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100%, to the nucleotide sequence as shown in SEQ ID NO: 1 or 4.

(57) In certain preferred embodiments, the nucleic acid molecule is a vector (for example, a cloning vector or an expression vector) comprising a genomic sequence or cDNA sequence of the PRV or modified form thereof as described herein, or a complementary sequence of the cDNA sequence. In certain preferred embodiments, the nucleic acid molecule is a vector (for example, a cloning vector or an expression vector) comprising a cDNA sequence of the PRV or modified form thereof as described herein, or a complementary sequence of the cDNA sequence.

(58) The nucleic acid molecule of the present invention can be delivered by any means known in the art, for example, by direct injection of naked nucleic acid molecule (e.g., naked RNA), or by using a non-viral delivery system. The non-viral delivery system can be prepared and obtained with various materials well known in the art, wherein the materials include but are not limited to those described in detail in “Yin H, et al. Nat Rev Genet. 2014 August; 15 (8): 541-55.” and “Riley M K, Vermerris W. Nanomaterials (Basel). 2017 Apr. 28; 7 (5). Pii: E94.”, all of which are incorporated herein by reference in their entirety, such as liposomes, inorganic nanoparticles (e.g., gold nanoparticles), polymers (e.g., PEG), etc.

(59) In certain preferred embodiments, the medicament comprises a therapeutically effective amount of the PRV and/or modified form thereof, or a therapeutically effective amount of the nucleic acid molecule as described herein. In certain preferred embodiments, the medicament may be in any form known in the medical arts. For example, the medicament may be tablet, pill, suspension, emulsion, solution, gel, capsule, powder, granule, elixir, lozenge, suppository, injection (including injection liquid, lyophilized powder), and other forms. In some embodiments, the medicament is an injection liquid or a lyophilized powder.

(60) In certain preferred embodiments, the medicament further comprises a pharmaceutically acceptable carrier or excipient. In certain preferred embodiments, the medicament comprises a stabilizer.

(61) In certain preferred embodiments, the medicament optionally further comprises an additional pharmaceutically active agent. In a preferred embodiment, the additional pharmaceutically active agent is a drug with anti-tumor activity, such as an additional oncolytic virus, chemotherapeutic

agent or immunotherapeutic agent.

(62) In the present invention, the additional oncolytic virus includes but is not limited to adenovirus, parvovirus, reovirus, Newcastle disease virus, vesicular stomatitis virus, measles virus, or any combination thereof. The chemotherapeutic agent includes but is not limited to 5-fluorouracil, mitomycin, methotrexate, hydroxyurea, cyclophosphamide, dacarbazine, mitoxantrone, anthracyclines (e.g., epirubicin or doxorubicin), etoposide, platinum compounds (e.g., carboplatin or cisplatin), taxanes (e.g., paclitaxel or taxotere), or any combination thereof. The immunotherapeutic agent includes but is not limited to immune checkpoint inhibitor (e.g., PD-L1/PD-1 inhibitor or CTLA-4 inhibitor), tumor-specific targeting antibody (e.g., rituximab or herceptin), or any combination thereof.

(63) In certain preferred embodiments, the medicament comprises a unit dose of the PRV and/or modified form thereof, for example, at least $1 \times 10^{2.2}$ pfu, at least $1 \times 10^{3.3}$ pfu, at least $1 \times 10^{4.4}$ pfu, $1 \times 10^{5.5}$ pfu, $1 \times 10^{6.6}$ pfu, at least $1 \times 10^{7.7}$ pfu, at least $1 \times 10^{8.8}$ pfu, at least $1 \times 10^{9.9}$ pfu, at least $1 \times 10^{10.10}$ pfu, at least $1 \times 10^{11.11}$ pfu, at least $1 \times 10^{12.12}$ pfu, at least $1 \times 10^{13.13}$ pfu, at least $1 \times 10^{14.14}$ pfu or at least $1 \times 10^{16.16}$ pfu of the PRV and/or modified form thereof. In certain preferred embodiments, the medicament comprises $1 \times 10^{2.2}$ pfu to $1 \times 10^{17.17}$ pfu of the PRV and/or modified form thereof.

(64) In certain preferred embodiments, the medicament comprises a unit dose of the nucleic acid molecule as described herein, for example, $1 \times 10^{10.10}$ to $5 \times 10^{14.14}$ (e.g., $3 \times 10^{14.14}$) virus genome copies of the nucleic acid molecule.

(65) In certain preferred embodiments, the medicament can be administered in combination with an additional therapy. This additional therapy may be any therapy known for tumors, such as surgery, chemotherapy, radiation therapy, immunotherapy, hormone therapy, or gene therapy. This additional therapy can be administered before, at the same time, or after administration of the medicament.

(66) In certain preferred embodiments, the tumor is selected from neuroglioma, neuroblastoma, gastric cancer, liver cancer, renal cancer, lung cancer, breast cancer, colon cancer, lymphoma, ovarian cancer, cervical cancer, endometrial cancer, melanoma, pancreatic cancer, osteosarcoma, prostate cancer, nasopharyngeal cancer, squamous cell carcinoma of nasal septum, larynx cancer, thyroid cancer, ductal carcinoma of thyroid, bladder cancer, etc. In certain exemplary embodiments, the tumor recurrence is recurrence of liver cancer.

(67) In certain preferred embodiments, the subject is a mammal, such as a human.

(68) In a second aspect, the present invention provides a method for treating a tumor and/or reducing or inhibiting a tumor recurrence, which comprises a step of administering to a subject in need thereof an effective amount of a PRV or modified form thereof, or an effective amount of a nucleic acid molecule; wherein, the nucleic acid molecule comprises a sequence selected from the following: (1) a genomic sequence or cDNA sequence of the PRV or modified form thereof; and (2) a complementary sequence of the cDNA sequence.

(69) In certain preferred embodiments, the subject is administered with the PRV. In certain preferred embodiments, the PRV is a wild-type PRV. In certain preferred embodiments, the PRV may be a strain isolated from an animal infected with a pseudorabies virus (PRV).

(70) In certain preferred embodiments, the genomic sequence of the PRV or modified form thereof has a sequence identity of at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100%, to the nucleotide sequence as shown in SEQ ID NO: 1. In certain preferred embodiments, the genomic sequence of the PRV or modified form thereof is the nucleotide sequence as shown in SEQ ID NO: 1.

(71) In certain preferred embodiments, the subject is administered with the modified form of the PRV. In certain preferred embodiments, the modified form is a modified PRV, which has a substitution, insertion or deletion of one or more nucleotides in the genome as compared to a wild-

type PRV.

(72) In certain preferred embodiments, the modified PRV has one or more modifications selected from the following as compared to a wild-type PRV: (1) a deletion or mutation of one or more endogenous genes; (2) a mutation, deletion or insertion of one or more nucleotides in an untranslated region (e.g., a promoter); (3) an insertion of one or more exogenous nucleotide sequences; and (4) any combination of the above three items.

(73) In certain preferred embodiments, the modified PRV does not express a functional EP0 protein.

(74) In certain preferred embodiments, the genome of the modified PRV of the present invention comprises the following modification: the EP0 gene comprises a loss-of-function mutation (e.g., an addition, deletion and/or substitution of one or more nucleotides), or is deleted or replaced with an exogenous nucleotide sequence (e.g., a nucleotide sequence encoding an exogenous protein). In certain preferred embodiments, the loss-of-function mutation is selected from missense mutation, nonsense mutation, frameshift mutation, base deletion, base substitution, base addition, and any combination thereof (e.g., deletion or substitution or addition of a gene fragment).

(75) In certain preferred embodiments, the EP0 gene is deleted. In such embodiments, the modified PRV does not express EP0 protein. In certain preferred embodiments, the EP0 gene is substituted with an exogenous nucleotide sequence (e.g., a nucleotide sequence encoding an exogenous protein).

(76) In certain preferred embodiments, the genomic sequence of the modified PRV has a sequence identity of at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% to the nucleotide sequence as shown in SEQ ID NO: 4. In certain preferred embodiments, the genomic sequence of the modified PRV is shown in SEQ ID NO: 4.

(77) In certain preferred embodiments, the modified PRV comprises a mutation, deletion or insertion of one or more nucleotides in a non-coding region (e.g., a promoter).

(78) In certain preferred embodiments, the genome of the modified PRV comprises the following modification: an original promoter of one or more PRV genes is substituted with a tumor-specific promoter. In the present invention, the original promoter of any PRV gene can be replaced with a tumor-specific promoter. In certain preferred embodiments, the tumor-specific promoter is selected from the group consisting of hTERT promoter, alpha-fetoprotein promoter, carcinoembryonic antigen promoter, prostate-specific antigen promoter, Survivin promoter and cyclooxygenase COX-2 promoter.

(79) In certain preferred embodiments, the modified PRV comprises an exogenous nucleotide sequence.

(80) In certain preferred embodiments, the exogenous nucleotide sequence encodes an exogenous protein selected from the group consisting of fluorescent protein, immunomodulatory polypeptide, cytokine, chemokine, anti-tumor protein or polypeptide.

(81) In certain preferred embodiments, the fluorescent protein is selected from the group consisting of green fluorescent protein, red fluorescent protein, blue fluorescent protein, yellow fluorescent protein, and any combination thereof.

(82) In certain preferred embodiments, the immunomodulatory polypeptide is selected from the group consisting of CD40L, OX40L, inducible costimulatory molecules (ICOS), FTL3L, LIGHT, CD137L, CD70, 4-1BB, GITR, CD28, and any combination thereof.

(83) In certain preferred embodiments, the cytokine is selected from the group consisting of interleukins (e.g., IL-2, IL-12 and IL-15), interferons (e.g., IFN α , IFN β , IFN γ), tumor necrosis factors (e.g., TNF α), colony stimulating factors (e.g., GM-CSF), and any combination thereof.

(84) In certain preferred embodiments, the chemokine is selected from the group consisting of CCL2, RANTES, CCL7, CCL9, CCL10, CCL12, CCL15, CCL19, CCL21, CCL20, XCL-1, and any combination thereof.

(85) In certain preferred embodiments, the anti-tumor protein or polypeptide is selected from the group consisting of: cytotoxic peptides, such as thymidine kinase TK (TK/GCV), TRAIL, FasL, or any combination thereof; immune checkpoint inhibitors, such as PD-1 antibody, anti-PD-L1 antibody, anti-TIGIT antibody, anti-BTLA antibody, anti-CTLA-4 antibody, anti-Tim-3 antibody, anti-Lag-3 antibody, anti-CD137 antibody, anti-OX40 antibody, anti-GITR antibody, anti-CD73 antibody, anti-KIR antibody, anti-ICOS antibody, anti-CSF1R antibody, or any combination thereof; tumor-specific targeting antibodies, such as anti-HER2 antibody (e.g., Herceptin), anti-CD20 antibody (e.g., rituximab), anti-VEGF antibody (e.g., bevacizumab), anti-EGFR antibody (e.g., cetuximab), or any combination thereof; protein or polypeptide that inhibits tumor angiogenesis, such as anti-vascular endothelial growth factor (anti-VEGF) antibody, VEGF-derived polypeptides (e.g., .sub.D(LPR), KSRVRKGKGQQRKRKRKKSRYK, etc.), ATN-161 or any combination thereof; and any combination of the above proteins or polypeptides.

(86) In certain preferred embodiments, the modified PRV comprises at least one insertion of the exogenous nucleotide sequence as described above and/or at least one of the mutation, deletion or insertion in an untranslated region (e.g., a promoter) as described above.

(87) In certain preferred embodiments, the modified PRV does not express a functional EP0 protein and comprises at least one insertion of the exogenous nucleotide sequence as described above and/or at least one of the mutation, deletion or insertion in an untranslated region (e.g., a promoter) as described above.

(88) In certain preferred embodiments, the PRV and modified form thereof as described above may be used in combination. Therefore, one or more of the PRV and modified forms thereof can be administered to the subject.

(89) In certain preferred embodiments, the nucleic acid molecule as described herein is administered to the subject.

(90) In certain preferred embodiments, the nucleic acid molecule consists of a genomic sequence or cDNA sequence of the PRV or modified form thereof as described herein, or a complementary sequence of the cDNA sequence.

(91) In certain preferred embodiments, the nucleic acid molecule has a genomic sequence of the PRV or modified form thereof as described herein. In certain preferred embodiments, the nucleic acid molecule has a nucleotide sequence selected from the following: (1) a nucleotide sequence as shown in SEQ ID NO: 1 or 4; (2) a nucleotide sequence having a sequence identity of at least 70%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100%, to the nucleotide sequence as shown in SEQ ID NO: 1 or 4.

(92) In certain preferred embodiments, the nucleic acid molecule is a vector (for example, a cloning vector or an expression vector) comprising a genomic sequence or cDNA sequence of the PRV or modified form thereof as described herein, or a complementary sequence of the cDNA sequence. In certain preferred embodiments, the nucleic acid molecule is a vector (for example, a cloning vector or an expression vector) comprising a cDNA sequence of the PRV or modified form thereof as described herein, or a complementary sequence of the cDNA sequence.

(93) In the present invention, the nucleic acid molecule as described herein can be delivered by any means known in the art, for example, by direct injection of naked nucleic acid molecule (e.g., naked RNA), or by using a non-viral delivery system. The non-viral delivery system can be prepared and obtained with various materials well known in the art, wherein the materials include but are not limited to those described in detail in “Yin H, et al. Nat Rev Genet. 2014 August; 15 (8): 541-55.” and “Riley M K, Vermerris W. Nanomaterials (Basel). 2017 Apr. 28; 7 (5). Pii: E94.”, all of which are incorporated herein by reference in their entirety, such as liposomes, inorganic nanoparticles (e.g., gold nanoparticles), polymers (e.g., PEG), etc.

(94) In certain preferred embodiments, the PRV and/or modified form thereof, or nucleic acid molecules as described herein, can be formulated and administered as a pharmaceutical

composition. Such pharmaceutical composition may comprise a therapeutically effective amount of the PRV and/or modified form thereof, or a therapeutically effective amount of the nucleic acid molecule as described herein. In certain preferred embodiments, the pharmaceutical composition may be in any form known in the medical arts. For example, the pharmaceutical composition may be a tablet, pill, suspension, emulsion, solution, gel, capsule, powder, granule, elixir, lozenge, suppository, injection (including injection liquid, lyophilized powder), and other forms. In some embodiments, the pharmaceutical composition is an injection liquid or a lyophilized powder.

(95) In certain preferred embodiments, the pharmaceutical composition further comprises a pharmaceutically acceptable carrier or excipient. In certain preferred embodiments, the pharmaceutical composition comprises a stabilizer.

(96) In the present invention, the PRV and/or modified forms thereof, or the nucleic acid molecules as described herein can be administered to a subject by various suitable means. In some cases, the route of administration of the PRV and/or modified form thereof or the nucleic acid molecule as described herein depends on the location and type of tumor. For example, for a solid tumor that is easily accessible, the virus or nucleic acid molecule is optionally administered by injection directly into the tumor (e.g., intratumoral injection); for a tumor of hematopoietic system, the virus or nucleic acid molecule can be administered by intravenous or other intravascular routes; for a tumor that is not easily accessible in the body (e.g., metastases), the virus or nucleic acid molecule can be administered systemically so that it can run over the whole body and thereby reaching the tumor (e.g., intravenous or intramuscular injection). Optionally, the virus or nucleic acid molecule of the present invention can be administered via subcutaneous, intraperitoneal, intrathecal (e.g., for brain tumors), topical (e.g., for melanoma), oral (e.g., for oral or esophageal cancer), intranasal or inhalation spray (e.g., for lung cancer) routes and so on. In certain preferred embodiments, the PRV and/or modified form thereof, or the nucleic acid molecule as described herein, can be administered via intradermal, subcutaneous, intramuscular, intravenous, oral routes etc.

(97) In certain preferred embodiments, the method further comprises administering an additional pharmaceutically active agent having anti-tumor activity. Such additional pharmaceutically active agent may be administered before, simultaneously or after administration of the PRV and/or modified form thereof, or the nucleic acid molecule as described herein.

(98) In certain preferred embodiments, the additional pharmaceutically active agent comprises additional oncolytic virus, chemotherapeutic agent, or immunotherapeutic agent.

(99) In the present invention, the additional oncolytic virus includes but is not limited to adenovirus, parvovirus, reovirus, Newcastle disease virus, vesicular stomatitis virus, measles virus, or any combination thereof. The chemotherapeutic agent includes but is not limited to 5-fluorouracil, mitomycin, methotrexate, hydroxyurea, cyclophosphamide, dacarbazine, mitoxantrone, anthracyclines (e.g., epirubicin or doxorubicin), etoposide, platinum compounds (e.g., carboplatin or cisplatin), taxanes (e.g., paclitaxel or taxotere), or any combination thereof. The immunotherapeutic agent includes but is not limited to immune checkpoint inhibitor (e.g., PD-L1/PD-1 inhibitor or CTLA-4 inhibitor), tumor-specific targeting antibody (e.g., rituximab or herceptin), or any combination thereof.

(100) In certain preferred embodiments, the PRV and/or modified form thereof can be administered in any amount of 1 to 1×10^{15} pfu/kg of the subject's body weight, for example, the PRV and/or modified form thereof can be administered in an amount of at least 1×10^3 pfu/kg, at least 1×10^4 pfu/kg, 1×10^5 pfu/kg, 1×10^6 pfu/kg, at least 1×10^7 pfu/kg, at least 1×10^8 pfu/kg, at least 1×10^9 pfu/kg, at least 1×10^{10} pfu/kg, at least 1×10^{11} pfu/kg, or at least 1×10^{12} pfu/kg of the subject's body weight. In certain preferred embodiments, the nucleic acid molecule as described herein can be administered in any amount of 1×10^{10} to 5×10^{14} (e.g. 3×10^{10} to 3×10^{14}) virus genome copies per kg of the subject's body weight. In certain preferred embodiments, the PRV and/or modified form thereof or the nucleic acid molecule as described herein can be administered 3 times per day, 2 times per day,

once per day, once every two days, or once per week, and the above-mentioned dosage regimen may be optionally repeated weekly or monthly as appropriate.

(101) In certain preferred embodiments, the method further comprises administering an additional therapy. This additional therapy may be any therapy known for tumors, such as surgery, chemotherapy, radiation therapy, immunotherapy, hormone therapy, or gene therapy. This additional therapy can be administered before, at the same time, or after administration of the method as described above.

(102) In certain preferred embodiments, the subject is a mammal, such as a human.

(103) In certain preferred embodiments, the tumor is selected from neuroglioma, neuroblastoma, gastric cancer, liver cancer, renal cancer, lung cancer, breast cancer, colon cancer, lymphoma, ovarian cancer, cervical cancer, endometrial cancer, melanoma, pancreatic cancer, osteosarcoma, prostate cancer, nasopharyngeal cancer, squamous cell carcinoma of nasal septum, larynx cancer, thyroid cancer, ductal carcinoma of thyroid, bladder cancer, etc. In certain exemplary embodiments, the tumor recurrence is recurrence of liver cancer.

(104) In a third aspect, the present invention also relates to a pharmaceutical composition, which comprises the PRV and/or modified form thereof as defined in the first or second aspect, or the nucleic acid molecule as defined in the first or second aspect.

(105) In certain preferred embodiments, the pharmaceutical composition may be in any form known in the medical art. For example, the pharmaceutical composition may be a tablet, pill, suspension, emulsion, solution, gel, capsule, powder, granule, elixir, lozenge, suppository, injection (including injection liquid, lyophilized powder), and other forms. In some embodiments, the pharmaceutical composition is an injection liquid or a lyophilized powder.

(106) In certain preferred embodiments, the pharmaceutical composition further comprises a pharmaceutically acceptable carrier or excipient. In certain preferred embodiments, the pharmaceutical composition comprises a stabilizer.

(107) In certain preferred embodiments, the pharmaceutical composition optionally further comprises an additional pharmaceutically active agent. In a preferred embodiment, the additional pharmaceutically active agent is a drug with anti-tumor activity, such as an additional oncolytic virus, chemotherapeutic agent or immunotherapeutic agent.

(108) In certain preferred embodiments, the pharmaceutical composition is used to treat a tumor and/or reduce or inhibit tumor recurrence in a subject.

(109) In certain preferred embodiments, the subject is a mammal, such as a human.

(110) In certain preferred embodiments, the tumor is selected from neuroglioma, neuroblastoma, gastric cancer, liver cancer, renal cancer, lung cancer, breast cancer, colon cancer, lymphoma, ovarian cancer, cervical cancer, endometrial cancer, melanoma, pancreatic cancer, osteosarcoma, prostate cancer, nasopharyngeal cancer, squamous cell carcinoma of nasal septum, larynx cancer, thyroid cancer, ductal carcinoma of thyroid, bladder cancer, etc. In certain exemplary embodiments, the tumor recurrence is recurrence of liver cancer.

(111) In a fourth aspect, the invention also relates to the PRV and/or modified form thereof as defined in the first or second aspect, or the nucleic acid molecule as defined in the first or second aspect, for use as a drug.

Beneficial Effects of Invention

(112) Compared with the prior art, the technical solution of the present invention has at least the following beneficial effects:

(113) The inventors of the present application have discovered for the first time that pseudorabies virus (PRV) has a broad-spectrum tumor-killing activity. Based on this finding, the present invention further provides an oncolytic virus based on PRV. The oncolytic virus of the present invention can be used alone in the treatment of a tumor, can also be used as an auxiliary method for traditional tumor treatment, or as a treatment method in the absence of other treatment methods, and thereby having great clinical value.

(114) The embodiments of the present invention will be described in detail below in conjunction with the drawings and examples, but those skilled in the art will understand that the following drawings and examples are only used to illustrate the present invention, not to limit the scope of the present invention. The various objects and advantageous aspects of the invention will become apparent to those skilled in the art from the following detailed description of the drawings and preferred embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A to 1C show the resultant micrographs of the in vitro killing experiments in Example 2 of wild-type PRV on human lung cancer cell line H1299, human liver cancer cell line BEL7402, human gastric cancer cell line BGC823, human colon cancer cell line HCT-116, mouse breast cancer cell line 4T1, human melanoma cell line MPWO, human cervical cancer cell line SIHA, mouse kidney cancer cell line Renca, human ovarian cancer cell line A2780, human nasopharyngeal cancer cell line CNE1, neuroglioma cell line GBM, human laryngeal cancer cell line Hep-2, human pancreatic cancer cell line Panc-1, human lymphoma cell line A20, mouse prostate cancer cell line Tramp C2, and human embryonic lung fibroblast MRCS, wherein MOCK indicates cells uninfected with the virus. The results show that the PRV showed a significant oncolytic effect on human and mouse tumor cell lines, but had less killing effect on MRCS of human non-tumor cell, after 72 hours of infection at a multiplicity of infection (MOI) of 1.

(2) FIG. 2 shows the killing effect of the wild-type PRV virus on mouse kidney cancer cell line Renca in Example 2. The results show that the Renca cells infected with PRV exhibited very obvious CPE in 24 hours, and they were almost all lysed to death by 48 hours.

(3) FIG. 3 shows the results of in vivo anti-tumor experiments in Example 3 of PRV-WT on human nasopharyngeal carcinoma model CNE1 (A), human Burkitt's lymphoma model Raji (B) and human neuroglioma model GBM (C). The results show that in the challenge groups, the growth of tumors formed by subcutaneous inoculation of CNE1, Raji or GBM cells in SCID mice significantly slowed down and arrested, and the tumors were even lysed and disappeared; in contrast, the tumors in the negative group (Mock) that were not treated with the oncolytic virus maintained normal growth, and their tumor volumes were significantly larger than those of the challenge groups.

(4) FIG. 4 shows the in vivo anti-tumor experiment results in Example 3 of PRV-WT on mouse colon cancer model CT26 (FIG. 4A), mouse liver cancer model Hep1-6 (FIG. 4B), mouse kidney cancer model Renca (FIG. 4C), mouse breast cancer model 4T1 (FIG. 4D). The results show that the PRV-WT had significant therapeutic effects in the above mouse tumor models.

(5) FIG. 5 shows the safety evaluation results of PRV-WT in the mouse intravenous injection model in Example 4. By intravenous injection of PBS (A) or $1 \times 10^{7.7}$ PFU virus (B) into Bab/c mice, the body weight and survival rate of the mice were monitored. The results show that the mouse body weight of the PRV-WT group and the PBS group showed the same trend, and none of the mice died, confirming that the wild-type PRV-WT had very good safety in the mouse intravenous model.

(6) FIG. 6 shows the safety evaluation results of wild-type PRV-WT in the mouse intracranial injection model in Example 4. By intracranial injection of $2 \times 10^{6.6}$, $2 \times 10^{5.5}$, and $2 \times 10^{4.4}$ PFU of PRV-WT into Bab/c mice, the survival rate of mice was monitored. The results show that the mice injected with the virus all died one after another, and showed a certain dose dependence. Such results suggest that the PRV-WT may have certain neurotoxicity.

(7) FIG. 7 shows the in vitro killing activity evaluation results of PRV-del-EP0 to tumor cell lines and diploid cell lines (similar to normal cell lines) in Example 5. FIG. 7A shows the killing results

of PRV-del-EP0 (BK61-dEP0) to various tumor cell lines, and FIG. 7B shows the killing results of PRV-del-EP0 (BK61-dEP0) to various diploid cell lines (similar to normal cell lines). The results show that PRV-del-EP0 had a tumor killing activity comparable to that of PRV-WT, and its killing activity to normal cells was reduced.

(8) FIG. 8 shows the safety evaluation results of PRV-del-EP0 in the mouse intracranial injection model in Example 5. By intracranial injection of $1 \times 10^{6.6}$, $1 \times 10^{5.5}$, $1 \times 10^{4.4}$, $1 \times 10^{3.3}$, $1 \times 10^{2.2}$, $1 \times 10^{1.1}$ PFU of PRV-WT (A) or PRV-del-EP0 (B) into mice, the survival rate of mouse was monitored. The results show that PRV-del-EP0 had significantly improved in vivo safety as compared to the wild-type PRV.

(9) FIGS. 9 to 10 show the therapeutic effects of PRV-del-EP0 on mouse liver cancer model in Example 5. Among them, FIG. 9 shows the effects of PRV-del-EP0 on the tumor size of the mouse liver cancer model; FIG. 10 shows the effects of PRV-del-EP0 on the survival rate of the mouse liver cancer model. The results show that PRV-del-EP0 had a significant antitumor activity comparable to that of PRV-WT.

(10) FIG. 11 shows the evaluation results of the tumor recurrence rate of the mice cured by PRV-WT and PRV-del-EP0 in Example 5. The results show that PRV-WT and PRV-del-EP0 could prevent tumor recurrence.

SEQUENCE INFORMATION

(11) The information of parts of sequences involved in the present invention is provided in Table 1 below.

(12) TABLE-US-00001 TABLE 1 Description of sequences SEQ ID NO: Description 1 Genome sequence of wild type PRV (PRV-WT) 2 Nucleic acid sequence encoding the early protein EP0 in the PRV genome 3 GFP gene sequence 4 Genome sequence of PRV-del-EP0

EXAMPLES

(13) The present invention will now be described with reference to the following examples intended to illustrate the invention (not to limit the invention).

(14) Unless otherwise specified, the molecular biology experimental methods and immunoassays used in the present invention basically referred to J. Sambrook et al., Molecular Cloning: A Laboratory Manual, 2nd Edition, Cold Spring Harbor Laboratory Press, 1989, and F M Ausubel et al., Short Protocols in Molecular Biology, 3rd Edition, John Wiley & Sons, Inc., 1995. The use of restriction enzymes was in accordance with the conditions recommended by the product manufacturers. If no specific conditions were indicated in the examples, the conventional conditions or the conditions recommended by the manufacturers should be followed. The used reagents or instruments, of which manufacturers were not given, were all conventional products that were commercially available. Those skilled in the art know that the examples describe the present invention by way of example, and are not intended to limit the claimed scope of the invention. All publications and other references mentioned herein are incorporated by reference in their entirety.

Example 1: Acquisition and Preparation of PRV and Modified Forms Thereof

(15) 1.1 Isolation of Pseudorabies Virus (PRV) from Specimens

(16) (1) Pharyngeal swabs and anal swabs of sick pigs were from Xiamen Center for Disease Control and Prevention, China, and pig embryonic kidney cells (PK-15; ATCC Number CCL-33™) were preserved by the National Engineering Research Center for Diagnostic Reagents and Vaccines for Infectious Diseases, Xiamen University, China, which were cultured in DMEM medium supplemented with 10% fetal bovine serum, glutamine, penicillin and streptomycin.

(17) (2) Treatment of specimens: the pharyngeal swabs and anal swabs were placed in a specimen preservation solution and fully stirred so as to wash off the virus and virus-containing cells attached to the swabs, and then the specimen preservation solution was subjected to high speed centrifugation under 4000 rpm and 4° C. for 30 min;

(18) (3) Inoculation and observation:

- (19) A. PK-15 cells were plated on a 24-well plate with 1×10^5 cells/well. The growth solution (DMEM medium, 10% fetal bovine serum, and glutamine, penicillin and streptomycin) was aspirated, and then each well was added with 1 mL of maintenance solution (DMEM medium, 2% fetal bovine serum, and glutamine, penicillin and streptomycin). Then except for the negative control wells, each well was inoculated with 50 μ L of sample supernatant, and cultured in an incubator at 37° C., 5% CO₂.
- (20) B. The cells were observed under microscope every day for one week and the occurrence of specific cytopathic effect (CPE) in the inoculated wells was recorded.
- (21) C. If the pseudorabies virus-specific CPE appeared in the cells of the inoculated wells within 7 days, the cells and supernatant were collected and cryopreserved at -80° C.; if no CPE appeared after 7 days, the cells were subjected to blind passage.
- (22) D. If CPE appeared in the cells within 6 blind passages, the cells and supernatant were collected and cryopreserved at -80° C.; if CPE did not appear after 6 blind passages, the cells were determined as negative.
- (23) (4) Virus isolation and cloning:
- (24) The viruses isolated from clinical specimens were identified by PCR, the pseudorabies virus-positive cultures were selected and subjected to at least 3 virus plaque purification experiments; the clonal strains obtained from virus plaques in each round were also identified by PCR, and pseudorabies virus-positive clonal strains were selected and subjected to the next round of cloning; and a single strain of pseudorabies virus with strong growth viability was selected as the candidate strain of oncolytic virus.
- (25) 1.2 Gene Editing of PRV Based on CRISPER/CAS9 Technology and Acquisition of the Modified Form Thereof
- (26) This example used wild-type PRV strain that was also referred to as BK61-WT (SEQ ID NO: 1) as an example to show how to obtain the PRV and modified form thereof for the present invention through gene editing. The specific method was as follows.
- (27) (1) sgRNA design based on CRISPER/CAS9 technology: the PAM site near the site where an exogenous gene was to be inserted was found by virus genome analysis, and the sgRNA was designed according to the sequence near the PAM, so that the CAS9 protein could cleave the virus genome, and then homologous recombination was performed to obtain a new modified virus.
- (28) (2) Construction of donor target fragments: the target fragments were sent to Shanghai Shengong for synthesis. The acquisition of target gene fragments were also the key to successful modification, when homologous recombination was performed.
- (29) Modified form: The gene sequence of key protein EP0 of wild-type PRV (of which the DNA sequence was shown in SEQ ID NO: 2) was replaced by the gene sequence of green fluorescent protein (GFP) (of which the DNA sequence was shown in SEQ ID NO: 3) to obtain the genome (of which the DNA sequence was shown in SEQ ID NO: 4) of the recombinant virus (which was named as PRV-del-EP0, and also called BK61-dEP0).
- (30) (3) The sgRNA constructed above was transferred into 293T cells to form a stable cell strain, then the target fragment was transferred into the cells, followed by infection with PRV virus. And a new recombinant virus was formed by using intracellular homologous recombination technology.
- (31) (4) The progeny virus formed after infection with 293T was used to infect PK-15 cells. The successfully recombined progeny virus carried a fluorescent signal and could be used for the screening and isolation of the progeny virus.

Example 2: In Vitro Evaluation of Anti-Tumor Activity of Wild-Type PRV

(32) 2.1 Virus and Cell Lines as Used

- (33) (1) Virus: In this example, the PRV-WT (SEQ ID NO: 1) provided in Example 1 was used.
- (34) (2) Cell lines: human rhabdomyosarcoma cell RD (ATCCR Number: CCL-136TM); human colorectal cancer cell lines SW1116 (ATCC® Number: CCL-233TM), SW480 (ATCC® Number: CCL-228TM) and HT-29 (ATCC® Number: HTB-38TM); human gastric cancer cell lines AGS

(ATCC® Number: CRL-1739TM), SGC7901 (CCTCC deposit number: GDC150), BGC823 (CCTCC deposit number: GDC151) and NCI-N87 (ATCC® Number: CRL-5822TM); human esophageal cancer cell line TE-1 (purchased from the Cell Resource Center, Shanghai Institute for Biological Sciences, Chinese Academy of Sciences, No. 3131C0001000700089); human small cell lung cancer cell line DMS114 (ATCC® Number: CRL-2066TM); human non-small cell lung cancer cell lines SPC-A-1 (CCTCC deposit number: GDC050), NCI-H1975 (ATCCR Number: CRL-5908TM), NCI-H1299 (ATCCR Number: CRL-5803TM), A549 (ATCCR Number: CCL-185TM), NCI-H661 (ATCCR Number: HTB-183TM), EBC-1 (purchased from Thermo Fisher Scientific, Inc., Catalog #: 11875101) and NCI-H1703 (ATCC® Number: CRL-5889TM); human liver cancer cell lines C3A (ATCC® Number: CRL-10741TM), HepG2 (ATCC® Number: HB-8065TM), SMMC7721 (purchased from the Basic Medical Cell Center, Institute of Basic Medical Sciences, Chinese Academy of Medical Sciences, Number: 3111C0001CCC000087), BEL7402 (CCTCC deposit number: GDC035), BEL7404 (purchased from the Cell Resource Center, Shanghai Institutes for Biological Sciences, Chinese Academy of Sciences, number: 3131C0001000700064), Huh7 (CCTCC deposit number: GDC134) and PLC/PRF/5 (ATCCR Number: CRL-8024TM); human ovarian cancer cell lines SKOV3 (ATCC® Number: HTB-77TM) and Caov3 (ATCC® Number: HTB-75TM); human endometrial cancer cell lines Hec-1-A (ATCC® Number: HTB-112TM), Hec-1-B (ATCC® Number: HTB-113TM) and Ishikawa (ECACC No. 99040201); human cervical cancer cell lines Hela (ATCC® Number: CCL-2TM), Caski (ATCC® Number: CRL-1550TM) and C-33A (ATCC® Number: HTB-31TM); human melanoma cell lines SK-MEL-1 (ATCC® Number: HTB-67TM) and MeWo (ATCC® Number: HTB-65TM); human breast cancer cell lines BcaP37 (CCTCC deposit number: GDC206), BT-474 (ATCC® Number: HTB-20TM) and MDA-MB-231 (ATCCR Number: HTB-26TM); human kidney cancer cell lines A-498 (ATCC® Number: HTB-44TM) and 786-0 (ATCCR Number: CRL-1932TM); human pancreatic cancer cell lines Capan-2 (ATCC® Number: HTB-80TM) and HPAF-2 (ATCC® Number: CRL-1997TM); human osteosarcoma cell line U2OS (ATCC® Number: HTB-96TM); human prostate cancer cell lines DU145 (ATCC® Number: HTB-81TM) and LNCap (ATCC® Number: CRL-1740TM); human neuroglioma cell line GBM (primary tumor cell line isolated from a patient tumor tissue); human neuroblastoma cell line SH-SY5Y (ATCC® Number: CRL-2266TM); human nasopharyngeal carcinoma cell line CNE (purchased from the Center for Basic Medical Cells, Institute of Basic Medical Sciences, Chinese Academy of Medical Sciences, No.: 3131C0001000700013); human nasal septum squamous cell carcinoma cell line RPMI 2650 (ATCC® Number: CCL-30TM); human laryngeal carcinoma cell line HEp-2 (ATCCR Number: CCL-23TM); human thyroid cancer cell line SW579 (preserved by the National Engineering Research Center for Diagnostic Reagents and Vaccines for Infectious Disease) and human ductal carcinoma of thyroid cell line TT (ATCC® Number: CRL-1803TM); human bladder cancer cell lines J82 (ATCCR Number: HTB-1TM) and 5637 (ATCC® Number: HTB-9TM); human Burkitt's lymphoma cell lines Daudi (ATCC® Number: CCL-213TM) and Raji (ATCC® Number: CCL-86TM); human normal cell lines including: human skin keratinocyte cell line HaCat (CCTCC, deposit number: GDC106), human embryonic lung fibroblast cell line MRC-5 (ATCC® Number: CCL-171TM), human foreskin fibroblast cell line HFF-1 (ATCC® Number: SCRC-1041TM), human prostate stromal cell line WPMY-1 (ATCCR Number: CRL-2854TM), human umbilical vein endothelial cell line HUVEC (purchased from Thermo Fisher Scientific, Inc., Catalog #: C01510C) and the differentiated human liver progenitor cell line HepaRG (with the characteristics of primary hepatocytes; purchased from Thermo Fisher Scientific, Inc., Catalog #: HPRGC10). The above cells were all preserved by the National Engineering Research Center for Diagnostic Reagents and Vaccines for Infectious Disease, Xiamen University, China. HepaRG cells were cultured in WME medium (added with 1.5% DMSO); AGS and TT were cultured in F-12K medium; SH-SY5Y was cultured in DMEM: F12 (1:1) medium; RD, C-33A, EBC-1, SK-MEL-1, J82 and DU145 were cultured in MEM medium; Raji, Daudi, 5637, 786-O, TE-1, Caski, NCI-

H1299, NCI-H1703, NCI-H1975, NCI-H661, SGC7901, BGC823, SW1116, HEP-2 and LNCap were cultured in RPMI-1640 medium; and other cells were cultured in DMEM medium. These mediums were all supplemented with 10% fetal bovine serum, glutamine and penicillin-streptomycin. All the above cells were cultured under standard conditions of 37° C. and 5% CO.sub.2.

(35) 2.2 Cultivation of Virus

(36) The RD cells were evenly plated on 10 cm cell culture plates, and the culturing conditions were DMEM medium containing 10% fetal bovine serum, glutamine, penicillin and streptomycin, 37° C., 5% CO.sub.2, and saturated humidity; when the cell confluence reached 90% or more, the cell culture medium was replaced with DMEM medium containing 2% serum, and each plate was inoculated with 10.sup.6 PFU of PRV-WT.

(37) After 24 hours of continuous culture, the PRV-WT proliferated in RD cells and caused CPE in the cells. When more than 90% of the cells turned contracted and rounded, showed increased graininess, and became detached and lysed, the cells and culture supernatant thereof were harvested. After freezing and thawing for 3 cycles, the culture supernatant was collected and centrifuged to remove cell debris, in which the centrifugation conditions were 4000 rpm, 10 min, and 4° C. Finally, the supernatant was filtered with 0.22 µm disposable filter (Millipore) to remove impurities such as cell debris.

(38) 2.3 Determination of Virus Titer

(39) The RD cells were plated on 6-well plates with cell density of 10.sup.5 cells/well. After the cells adhered, the virus was diluted 10-fold. 100 µl of the dilution of virus was added to each well to infect the cells, followed by shaking and mixing once every 15 minutes. After shaking and mixing 5 times, the supernatant was removed. 2% agarose solution prepared with pbs was dissolved by heating and then mixed with DMEM medium containing 10% serum at a volume ratio of 1:1, and was added to the cells. When it was cooled and solidified, it was inverted and placed in an incubator. After cultivating for three days, 10% formaldehyde solution was added for fixation for 1 hour, and then the gel was inverted and taken out, followed by staining with crystal violet staining solution for 15 minutes. The titer of the virus was determined by counting the number of plaques formed.

(40) 2.4 In Vitro Anti-Tumor Experiments of Virus

(41) The human tumor cells and normal cells were inoculated into 96-well plates at 10.sup.4 per well. After the cells adhered, each well was replaced with corresponding cell culture medium without serum, and inoculated with viruses with MOIs of 10, 1, 0.1 and 0.01, respectively. Subsequently, CPE of the cells were monitored daily by a microscope.

(42) The micrographs are shown in FIGS. 1A to 1C. The results show that after 72 hours of infection with a multiplicity of infection (MOI) of 1, a significant reduction in the number of the tumor cells, marked shrinking and lysis and the like, were detected in the virus-infected groups; while the non-tumor cells infected with the virus had less change in cell morphology as compared to the non-tumor cells in the Mock group. The above results indicate that the PRV has a significant oncolytic effect on a variety of human and murine tumor cell lines, but has little effect on the non-tumor cell human embryonic lung fibroblast MRCS. In addition, after 24 hours of infection with a MOI of 1, the CPE of Renca cells was very obvious, and almost all of the cells were lysed to death by 48 hours (FIG. 2).

(43) Cell Counting Kit-8 (CCK-8 kit; Shanghai Biyuntian Biotechnology Co., Ltd.) was used to detect cell survival rate after 72 hours of virus infection and culture. The specific methods were as follows:

(44) For adherent cells, the original medium in a 96-well cell culture plate was directly discarded; for suspension cells, the original medium in a 96-well cell culture plate was carefully discarded after centrifugation; and then 100 µl of fresh serum-free medium was added per well. 10 µl of CCK-8 solution was added to each of the wells inoculated with cells, and an equal amount of CCK-

8 solution was also added to the blank culture medium as a negative control, followed by incubation at 37° C. in a cell culture incubator for 0.5-3 hours. The absorbance was detected at 450 nm using a microplate reader at 0.5, 1, 2, 3 hours, respectively, and the time point where the absorbance was within a suitable range was selected as a reference for cell survival rate. The CCK-8 detection results of the cells against PRV-WT are shown in Table 2, where “-” indicated that the cell survival rate after virus treatment was not significantly different from that of the MOCK group; “+” indicated that after virus treatment, the cell number was reduced, the survival rate was still greater than 50% but was significantly different from that of the MOCK group; “++” indicated that the cell survival rate after virus treatment was less than 50%, and was significantly different from that of the MOCK group.

(45) The calculation method of cell survival rate is:

$$(46) \text{ survival_rate(\%)} = \frac{\frac{(\text{read_of_test_group} - \text{read_of_negative_control_group})}{(\text{read_of_positive_control_group} - \text{read_of_negative_control_group})}}{\times 100\%}$$

(47) TABLE-US-00002 TABLE 2 Results of in vitro anti-tumor test of PRV-WT MOI Cell line 10
1 0.1 0.01 A549 ++ ++ ++ + H661 ++ ++ ++ + H1299 ++ ++ ++ ++ LLC ++ + - - BEL7704 ++
++ ++ ++ BEL7402 ++ ++ ++ ++ GSQ7701 ++ ++ ++ + HEP1-6 ++ ++ ++ + HUH7 ++ ++ ++ ++
QGY7703 ++ + + - SMMC7721 ++ ++ ++ + AGS ++ ++ ++ ++ BGC823 ++ + + + SGC7901 ++
++ ++ ++ CT26 ++ ++ ++ + HCT116 + - - - SW1116 ++ + + + MCF7 ++ ++ ++ + 4T1 ++ + + +
MEWO ++ ++ ++ ++ B16 ++ ++ + - Renca ++ ++ ++ + SKOV3 ++ ++ ++ ++ A2780 ++ ++ ++
++ CNE1 ++ ++ ++ ++ CNE2 ++ ++ ++ + GBM ++ ++ ++ ++ Hep-2 ++ ++ ++ + Panc-1 ++ ++ ++
++ Raji ++ ++ ++ + A20 ++ ++ ++ ++ Tramp C2 ++ ++ ++ ++ MRC5 ++ + - - Note: “-” indicated
that there was no significant difference in cell survival rate between virus treatment group and
MOCK group; “+” indicated that after virus treatment, the number of cells was reduced, the
survival rate was greater than 50% but was significantly different from that of MOCK group; “++”
indicated that the cell survival rate after virus treatment was less than 50%, and was significantly
different from that of the MOCK group.

(48) It can be seen from Table 2 that the PRV-WT had a good killing effect on most of the detected tumor cells. In particular, the virus had a very significant killing effect on lung cancer, liver cancer, ovarian cancer, neuroblastoma, cervical cancer, lymphoma, and kidney cancer. On the other hand, the PRV-WT had a certain killing effect on non-tumor cell lines including human embryonic lung fibroblast cell line MRC-5.

Example 3: In Vivo Anti-Tumor Experiment of Wild-Type PRV

(49) 3.1 Virus, Cell Lines and Experimental Animals

(50) (1) Virus: In this example, the PRV-WT provided in Example 1 was used. For the virus cultivation and virus titer determination methods, see Examples 2.2 and 2.3, respectively.
(51) (2) Cell lines: human nasopharyngeal carcinoma cell line CNE1, human Burkitt's lymphoma cell Raji (ATCC® Number: CCL-86™), human neuroglioma cell line GBM (primary tumor cell line isolated from patient tumor tissue), mouse colon cancer cell CT26, mouse liver cancer cell Hep1-6, mouse kidney cancer cell Renca and mouse breast cancer cell 4T1. Except Raji, the above cells were cultured in RPMI-1640 medium, and all other cells were cultured in DMEM medium; and the above mediums were all added with 10% fetal bovine serum, glutamine and penicillin-streptomycin. All the above cells were cultured under standard conditions of 37° C. and 5% CO.sub.2.

(52) (3) Experimental animals: 6-8 week-old female C.B17 SCID mice or Bab/c mice were from Shanghai Silaike Experimental Animal Co., Ltd.; according to the plan approved by Experimental Animal Center and Ethics Committee, Xiamen University, the mice were raised under SPF conditions.

(53) 3.2 In Vivo Anti-Tumor Experiments of Virus

(54) For the human tumor transplantation model, SCID mice were used. The tumor cells used for subcutaneous tumor formation were digested with 0.01% trypsin, and then resuspended into a single cell suspension using cell culture medium containing 10% fetal bovine serum. The cell density of the suspension was counted. The cells were precipitated by centrifugation under 1000 g for 3 min, and then the cells were resuspended with an appropriate volume of PBS to reach a concentration of about 10×10^6 – 10×10^7 cells/100 μ l PBS. The tumor cells were subcutaneously inoculated in the back of SCID mice at 10×10^6 – 10×10^7 cells/100 μ l PBS/site with a syringe. When the tumor cells grew into a tumor mass of about 100 mm³ under the skin of SCID mice after about 14–21 days, the tumor-bearing SCID mice were randomly divided into experimental groups treated with PRV-WT (BK61) and the negative control group (Mock). For the mouse tumor model, Bab/c mice were used, and were subcutaneously inoculated with tumor cells. After 7–14 days, the mice with tumor mass of about 100 mm³ were selected for treatment.

(55) The oncolytic virus PRV-WT at a dose of 10×10^6 TCID₅₀/100 μ l serum-free medium/tumor mass and an equal amount of serum-free medium were intratumorally injected respectively, once every two days for a total of 5 times of treatment.

(56) The tumor size was measured with a vernier caliper and recorded every two days. The calculation method of tumor size is:

Tumor size (mm³)=tumor length value $\times$ (tumor width value)²/2.

(57) The treatment results of PRV-WT on human tumor transplantation model and mouse tumor model are shown in FIG. 3 and FIG. 4, respectively. The results show that after treatment with PRV-WT, the growth of the tumors of CNE1 (FIG. 3A), GBM (FIG. 3B), Raji (FIG. 3C) and CT26 (FIG. 4A), Hep1-6 (FIG. 4B), Renca (FIG. 4C) and 4T1 (FIG. 4D) gradually slowed down and arrested, and the tumors were even lysed and disappeared; in contrast, the tumors in the negative group (Mock) maintained normal growth, and the tumor sizes were significantly larger than those in the experimental groups.

(58) The above results indicate that PRV-WT exhibited significantly favorable antitumor activity in vivo.

Example 4: Safety Evaluation of Oncolytic Virus

(59) 4.1 Virus and Experimental Animals as Used

(60) (1) Virus: In this example, the PRV-WT provided in Example 1 was used. For virus cultivation and virus titer determination methods, Examples 2.2 and 2.3 were referred to, respectively.

(61) (2) Experimental animals: 6–8 weeks old Bab/c mice were from Shanghai Silaike Experimental Animal Co., Ltd.; according to the protocol approved by the Experimental Animal Center and Ethics Committee, Xiamen University, the mice were raised under clean-grade conditions, and subsequently used for in vivo virulence assessment of pseudorabies virus.

(62) 4.2 Safety Evaluation of Virus in Mice

(63) (1) Bab/c mice were selected and subjected to single intravenous injection of PRV-WT or PBS, the challenge titer dose was 10×10^7 TCID₅₀/mouse (6 mice per group), and then the survival rate and body weight of the Bab/c mice of the challenge group were monitored and recorded every day. The statistical results of body weight of the mice after injection of PBS or PRV-WT are shown in FIGS. 5A–B, which showed that within 15 days after challenge, none of the mice in the challenge group died, and the trend of animal body weight growth of the challenge group was consistent with that of the control group, i.e., there was no statistical difference ($P>0.05$). This result indicates that PRV-WT had very good safety in the mouse intravenous model.

(64) (2) Bab/c mice were selected and intracranially injected with different doses of PRV-WT with challenge titers of 2×10^6 , 2×10^5 , 2×10^4 PFU/mouse (4 mice per group). Subsequently, the survival rates of Bab/c mice in different dose challenge groups were recorded every day. The results are shown in FIG. 6, which showed that the death of mice occurred one after another, and this phenomenon was dose-dependent, indicating that PRV-WT had certain neurotoxicity.

Example 5: Anti-Tumor Activity and Safety Evaluation of Modified Form of PRV

(65) 5.1 Virus as Used:

(66) In this example, the PRV-del-EP0 (SEQ ID NO: 4) provided in Example 1 was used. For virus cultivation and virus titer determination methods, Examples 2.2 and 2.3 were referred to, respectively.

(67) 5.2 In Vitro Oncolytic Activity Evaluation of PRV-Del-EP0

(68) The target cells were treated with PRV-del-EP0 (BK61-dEP0) with MOI=1 according to the method described in Example 2, and the survival rate of cells after treatment with PRV-del-EP0 (BK61-dEP0) was detected using CCK8 method. FIG. 7A shows the killing results of PRV-del-EP0 (BK61-dEP0) on various tumor cell lines, and the results indicate that it substantially maintained a killing effect comparable to that of wild-type PRV of parental strain. FIG. 7B shows the killing results of PRV-del-EP0 (BK61-dEP0) on a variety of non-tumor cell lines, and the results indicate that PRV-del-EP0 showed a significantly reduced killing activity to diploid cell lines (similar to normal cell line) as compared to the wild-type PRV of parent strain. The above results indicate that PRV-del-EP0 not only retained the significant tumor-killing activity of wild-type PRV, but also had an improved safety to some extent.

(69) 5.3 In Vivo Safety Evaluation of PRV-Del-EP0

(70) ICR mice were selected and intracranially injected with different doses of PRV-WT and PRV-del-EP0 with challenge titer doses of $1 \times 10^{6.6}$, $1 \times 10^{5.5}$, $1 \times 10^{4.4}$, $1 \times 10^{3.3}$, $1 \times 10^{2.2}$, $1 \times 10^{1.1}$ PFU/mouse (4 mice per group), and then the survival rates of ICR mice in different dose challenge groups were recorded every day. The results of the PRV-WT (BK61-WT) group are shown in FIG. 8A, and the results of the PRV-del-EP0 (BK61-dEP0) group are shown in FIG. 8B. The results show that the mice in the PRV-WT group died one after another, all the mice in the $1 \times 10^{6.6}$ and $1 \times 10^{5.5}$ groups died, and the survival rate of the mice in the $1 \times 10^{4.4}$ PFU group was only 25%; in contrast, among the PRV-del-EP0 groups, the death of mice only occurred in the $1 \times 10^{6.6}$ group, and the survival rate was 50%. The above results indicate that the PRV-del-EP0 had significantly improved in vivo safety as compared to the wild-type PRV.

(71) 5.4 Evaluation of In Vivo Therapeutic Effect of PRV-del-EP0

(72) The tumor cells Hep1-6 used for subcutaneous tumor formation in C57/B6 immune mice were digested with 0.01% trypsin, and then resuspended into a single cell suspension using cell culture medium containing 10% fetal bovine serum. The cell density of the suspension was counted. The cells were precipitated by centrifugation under 1000 g for 3 min, and then the cells were resuspended with an appropriate volume of PBS to reach a concentration of about $10^{6.6}$ - $10^{7.7}$ cells/100 μ l PBS. The tumor cells were subcutaneously inoculated in the back of C57/B6 mice at $10^{6.6}$ - $10^{7.7}$ cells/100 μ l PBS/site with a syringe. When the tumor cells grew into a tumor mass of about $10^{3.3}$ mm under the skin of SCID mice after about 7-14 days, the tumor-bearing SCID mice were randomly divided into 3 groups, which were intratumorally injected with PRV-WT (BK61-WT), PRV-del-EP0 (BK61-dEP0) and PBS, respectively, once every two days, for a total of 3 treatments. The tumor size was measured with a vernier caliper and recorded every two days, and the method for calculating the tumor size was:

$$\text{Tumor size (mm}^3\text{)} = \text{tumor length value} \times (\text{tumor width value})^2 / 2.$$

(73) The results are shown in FIG. 9. The tumors were completely cleared in the mice of the PRV-WT group and the PRV-del-EP0 group, indicating that the PRV-del-EP0 exhibited the same significant in vivo therapeutic effect as the wild strain.

(74) At the same time, the survival rates of the mice were determined. According to animal ethics, the mice were killed when the tumor size reached 2000 mm^3 . The results are shown in FIG. 10, which indicate that both PRV-WT and PRV-del-EP0 treatments could significantly improve the survival rate of the mice.

(75) Further, the tumor recurrence rate in the mice was evaluated. Specifically, the mice that were cured in the above-mentioned PRV-WT group and PRV-del-EP0 group were inoculated again with

tumors, and the number of inoculated cells was 10 times the number of the initially inoculated cells; and, the mice that were not treated with PRV-WT and PRV-del-EP0 were used as the control group (NC) and inoculated with the same amount of tumor cells. The tumor growth of the mice was monitored, the number of the mice with recurring tumors was counted, and the tumor recurrence rate was calculated as the following: tumor recurrence rate=(number of mice with recurring tumor/total number of mice inoculated with tumor)×100%. The results are shown in FIG. 11, which showed that none of the mice in the PRV-WT group and the PRV-del-EP0 group had recurring tumors, while tumors were observed in all of the mice of the control group that had not been treated with PRV-WT and PRV-del-EP0. This result suggests that mice cured by PRV-WT and PRV-del-EP0 had good anti-tumor immunity and tumor recurrence could be prevented.

(76) Although the specific embodiments of the present invention have been described in detail, those skilled in the art will understand that various modifications and changes can be made to the details based on all the teachings that have been published, and these changes are within the protection scope of the present invention. The protection scope of the invention is given by the appended claims and any equivalents thereof.

Claims

1. A method for treating a tumor in a subject, the method comprising: administering to a subject in need thereof, an effective amount of a modified pseudorabies virus (PRV), or a nucleic acid molecule; wherein the nucleic acid molecule comprises a nucleotide sequence selected from the group consisting of: (1) a genomic sequence or cDNA sequence of the modified PRV; and (2) a complementary sequence of the cDNA sequence, and wherein the modified PRV does not express a functional EP0 protein compared to a wild-type PRV, and wherein the tumor is selected from the group consisting of neuroglioma.
2. The method of claim 1, wherein the genomic sequence of the wild-type PRV has a sequence identity of at least 70% to the nucleotide sequence of SEQ ID NO: 1.
3. The method of claim 1, wherein the genome of the modified PRV comprises the following modification: an original promoter of at least one PRV gene is replaced with a tumor-specific promoter.
4. The method of claim 1, wherein the modified PRV comprises an exogenous nucleotide sequence; and wherein the exogenous nucleotide sequence encodes an exogenous protein selected from the group consisting of a fluorescent protein, immunomodulatory polypeptide, cytokine, chemokine, and anti-tumor protein or polypeptide.
5. The method of claim 1, wherein the nucleic acid molecule has a genomic sequence of the modified PRV.
6. The method of claim 1, wherein the nucleic acid molecule is a vector comprising a cDNA sequence of the modified PRV or a complementary sequence of the cDNA sequence.
7. The method of claim 1, further comprising administering at least one additional pharmaceutically active agent having an anti-tumor activity to the subject.
8. The method of claim 1, wherein the subject is a human.
9. The method of claim 1, wherein a genome of the modified PRV comprises the following modification: (i) an EP0 gene comprising a loss-of-function mutation, or (ii) an EP0 gene which is deleted or substituted with an exogenous nucleotide sequence encoding an exogenous protein.
10. The method of claim 9, wherein the loss-of-function mutation is at least one selected from the group consisting of a missense mutation, nonsense mutation, frameshift mutation, base deletion, base substitution, base addition, and a combination thereof.
11. The method of claim 1, wherein a genomic sequence of the modified PRV has a sequence identity of at least 70% to the nucleotide sequence of SEQ ID NO: 4.
12. The method of claim 7, wherein the additional pharmaceutically active agent is at least one

selected from the group consisting of an oncolytic virus, chemotherapeutic agent and immunotherapeutic agent.

13. The method of claim 12, comprising at least one selected from the group consisting of the following (i), (ii), and (iii): (i) the additional oncolytic virus is selected from the group consisting of adenovirus, parvovirus, reovirus, Newcastle disease virus, vesicular stomatitis virus, measles virus, and a combination thereof; (ii) the chemotherapeutic agent is selected from the group consisting of 5-fluorouracil, mitomycin, methotrexate, hydroxyurea, cyclophosphamide, dacarbazine, mitoxantrone, anthracyclines, etoposide, platinum compounds, taxanes, and a combination thereof; and (iii) the immunotherapeutic agent is selected from the group consisting of an immune checkpoint inhibitor, a tumor-specific targeting antibody, and a combination thereof.

14. A method for reducing or inhibiting tumor recurrence in a subject, the method comprising: administering, to a subject in need thereof, an effective amount of a wild-type pseudorabies virus (PRV), or a modified PRV, or a nucleic acid molecule; wherein the nucleic acid molecule comprises a sequence selected from the group consisting of: (1) a genomic sequence or cDNA sequence of the wild-type PRV or the modified PRV; (2) a complementary sequence of the cDNA sequence; and wherein the modified PRV does not express a functional EP0 protein compared to the wild-type PRV, and wherein the tumor is selected from the group consisting of neuroglioma.

15. The method of claim 14, wherein a genomic sequence of the wild-type PRV has a sequence identity of at least 70% to the nucleotide sequence of SEQ ID NO: 1.

16. The method of claim 14, wherein a genome of the modified PRV comprises the following modification: (i) an EP0 gene comprising a loss-of-function mutation, or (ii) an EP0 gene which is deleted or substituted with an exogenous nucleotide sequence encoding an exogenous protein.

17. The method of claim 14, wherein a genomic sequence of the modified PRV has a sequence identity of at least 70% to the nucleotide sequence of SEQ ID NO: 4.

18. The method of claim 14, wherein the nucleic acid molecule has a genomic sequence of the wild-type PRV or the modified PRV; or the nucleic acid molecule is a vector comprising a cDNA sequence of the wild-type PRV or the modified PRV, or a complementary sequence of the cDNA sequence.
